Health Insurance Policies

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PART 1

Part a. As we need to model claim frequency we will use a discrete distribution. As claims is a rare event, data is collected over a set period and claims are independent Poisson is theoretically appropriate.

As poisson distribution has a key assumption of equidispersion we can test the mean-variance ratio which will ideally be close to 1. For chronic conditions, the variance to mean ratio is 0.95. For non-chronic conditions the variance to mean ratio is 0.96. As both of these values are within 5% and under 1, it suggest slight under-dispersion but is suitable. Therefore, I will use the Poisson distribution where $\lambda = 0.07852$ for chronic conditions and $\lambda = 0.05658$ for non-chronic conditions. Alternatives could have been negative binomial but this is more suitable for over-dispersion and binomial was not suitable as multiple claims couldn't be measured.

Using a confidence level of 95% with a normal approximation produces the following results:

Confidence Levels		
Chronic Conditions	**Lower CI**	**Upper CI**
Yes	0.06121	0.09585
No	0.04921	0.06396

Part b. The data should be interpreted cautiously because of the under-dispersion Poisson variance assumption is only semi-satisfied. However, given the small deviation Poisson is a reasonable approximation. Results should acknowledge the limitations of using this model.

PART 2

Firstly, we can observe the claim amounts across the age group bins which were given to us: 20-39, 40-59 and 60+.
There were 3 main findings:
1: 40-59 chronic group has the highest mean and median amounts.
2: Claim rate increases over both age and chronic conditions.
3: 60+ non-chronic has the highest claim number.

For claim rates which is a binary indicator we use a chi-square test to test across age groups and chronic condition. This reveals that age and chronic condition are both statistically significant with p-values of 1.32×10^{-9} and 0.013 respectively. For claim amounts which is continuous we will use an anova test as for the 3 age groups and a t-test for chronic condition. This reveals age is not significant ($p = 0.55$) but chronic condition is significant ($p = 0.021$).
These results suggest age should remain a core factor for modelling claim rates and age-based premiums should be utilised. Chronic status is a compound risk affecting both frequency and

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amounts. Underwriting and pricing should reflect this by applying stronger differentiation and subgrouping for chronic conditions.

PART 3

For smokers 8.87% file claims and for non-smokers 5.67% file claims. This is a 56.4% relative increase in claim frequency if someone smokes. The average claim amount for smokers is $83623.1 and $80867.35 for non-smokers. This is a 3.4% relative increase in the claim severity. For non-smokers with non-chronic illnesses their claim rate is 5.23% which is our baseline. If they have chronic illness this increases by 2.21% to 7.44%. For a smoker with a non-chronic illness, their claim rate is 8.55% which increases to 10.08% a 4.86% increase if they have a chronic illness.

Chi-square for claim frequency reveals that smoking is significant with a p-value of 0.0033. The t-test for claim-severity reveals that smoking is not-significant with a p-value of 0.7636. Smoking does not amplify the effect of chronic illness. The marginal effect of chronic effect is smaller among smokers (1.53%) than non-smokers (2.21%) indicating sub-additive interactions.

PART 4

1. Selection Bias
As the data is from Critical Illness policies and customers choose to purchase this there is an inclination for them to have an illness before they purchase the policy. Therefore, these types of people may be over-represented.

2. Insufficient granularity
There are only three age bands, (20-39, 40-59, 60+) which limits the ability to detect finer trends. Chronic conditions and smoking are binary indicators, concealing severity level, duration and intensity. Additionally, claim characteristics such as timing and complexity are missing. Furthermore, only 302 out of the 5000 people filed claims (6.04%) reducing statistical power.

PART 5

We can start by creating the hypothesis test defined below to determine the minimum sample size.

$$H_0\colon \mu_{\text{chronic claim amount}} = \mu_{\text{non chronic claim amount}}$$
$$H_1\colon \mu_{\text{chronic claim amounts}} > \mu_{\text{non chronic claim amount}}$$

Using the two sample difference in means formula with 95% confidence ($\alpha = 0.05$) and 80% power ($\beta = 0.2$).

$$n = \frac{2(z_{1-\alpha} + z_{1-\beta})^2 \times \sigma^2}{(\mu_1 - \mu_2)^2}$$

Calculating n to be 116.72 we can deduce a minimum of 117 claimants from both sides assuming a 80% power and a 95% confidence level to reach 95% certainty. As there are only 78 current chronic claimants a further 39 are required. As there are 224 non-chronic claimants this is sufficient.

Overall, analysis confirms chronic conditions represents a significant compound risk justifying differential pricing. While consumer protection concerns are valid, risk-based pricing reflects claim patterns to ensure fairness to all policyholders.